(b) greater resultant (downward opposing) force, so lesser height [A1]

5 (a) No resultant (external) force (acts on the car-truck system) so (by principle of conservation of momentum) the total momentum is conserved. [A1]

(b) conservation of momentum: $\rightarrow$ (0.800)(9.2) + 0 = (3.200) V $\Rightarrow$ V = 2.3 m s^{-1} [C1]

initial kinetic energy = $\frac{1}{2}mv^2 = \frac{1}{2}(0.800)(9.2)^2 = 33.856$ J [A1]

final kinetic energy = $\frac{1}{2}mv^2 = \frac{1}{2}(3.200)(2.3)^2 = 8.464$ J

percentage loss in kinetic energy = $\frac{33.856 - 8.464}{33.856} \times 100\% = 75\%$

(c) conservation of energy: $\frac{1}{2}m_{A\&B}v^2 = \frac{1}{2}kx^2$

$\frac{1}{2}(3.2)(2.3)^2 = \frac{1}{2}(2500)x^2$ [M1]

$x = 0.082$ m [A1]

6 (a) Straight lines in uniform radial pattern centred on charge [B1]